When does dawn become visible? The Quranic dawn (fajr al-sadiq) and dusk (shafaq) as a detection problem, tested against 116 years of observation

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Abstract

Islamic prayer timekeeping requires the observation, or prediction, of four twilight events: the true dawn (fajr al-sadiq, the white thread of light spreading along the horizon), the false dawn that precedes it (fajr al-kadhib, the zodiacal wolf's tail), and the evening disappearance of the red and of the white twilight (shafaq al-ahmar and al-abyad). Modern practice replaces these observations with fixed solar depression angles, and the major conventions disagree by tens of minutes because a fixed angle cannot represent what the sources describe: a visual detection event, dependent on atmosphere, background sky, and the human observer. We formulate the dawn criterion of Quran 2:187 as a quantitative psychophysical detection problem: simulated absolute sky radiance across a horizon band of patches, mesopic photometry, and a Weber contrast test of each patch's growth above its own deep-night baseline, requiring simultaneous distinctness across the band's lateral extent. Sky radiance is supplied by a validated backward Monte Carlo radiative transfer model (companion paper); the detection layer contains a single calibrated constant bridging laboratory contrast thresholds to field detection of a borderless gradient. Calibrated on the desert campaigns that report a documented central-mean dawn depression, the model reproduces sixteen published visual and instrumental campaigns spanning 116 years, latitudes 6°S to 52°N, and desert, agricultural, coastal, and urban backgrounds, with a median absolute residual of 0.26° of solar depression over the fourteen scored rows. An out-of-sample year-long panel study at Birmingham, UK, is reproduced with zero retuning (mean residual +0.54°, RMS 0.95° over all 42 published dates), including its double-peaked seasonal curve and its summer relaxation to about 12.5°; on the same benchmark the fixed angles in worldwide use miss by 1.8 to 4.4° RMS, and the most widely used one is undefined on 20 of the 42 dates. The calibrated constant inverts consistently ($f = 51$ to 57) across every campaign that documents a central mean, with no latitude trend: it behaves as a single property of human dawn detection rather than a fit artifact. We further compute the false dawn as a distinct earlier detection of the zodiacal column, reproduce the qualitative suppression of its distinctness by urban skyglow, and quantify how measured cloud fields, aerosol, and artificial light shift the times, with propagated uncertainties reported per event. Computed times are offered as a research instrument, not a replacement for established calendars.

Keywords: fajr, twilight, prayer times, visual detection, contrast threshold, zodiacal light, sky brightness

1. Introduction

The Islamic day is structured by observable celestial events, and two of them are twilight phenomena of considerable optical subtlety. The Quran defines the onset of the daily fast, which coincides with the earliest time of the dawn prayer, by a visual criterion: "eat and drink until the white thread of dawn becomes distinct to you from the black thread [of night]" (Quran 2:187). Prophetic tradition sharpens the geometry: the true dawn (al-fajr al-sadiq) "spreads laterally along the horizon," while the false dawn that precedes it (al-fajr al-kadhib) rises "vertically, like the tail of the wolf," and does not begin the fast. The evening prayer window is likewise bounded by the disappearance of twilight (al-shafaq), with the classical schools differing over whether the red glow or the later white glow is decisive.

Contemporary practice almost universally replaces these observations with fixed solar depression angles: 18° (Muslim World League), 15° (ISNA), 19.5° (Egyptian General Authority), or fixed clock offsets (Umm al-Qura). The conventions disagree with one another by tens of minutes at mid-latitudes and diverge unboundedly toward the poles, where some depression angles are never reached in summer. The disagreement is structural rather than empirical: a fixed angle asserts that dawn visibility is a function of solar geometry alone, whereas the defining texts describe a detection event, which necessarily depends on the radiance of the dawn band, the darkness of the adjacent sky, the state of the atmosphere, and the human visual system. Observation campaigns confirm the structural point: measured naked-eye dawn depressions cluster near 14 to 15° under desert skies [1, 9, 10], sit lower under urban or humid skies, and drift with season at high latitude [12], none of which a constant can express.

This paper takes the textual criterion seriously as a physics problem. We formulate "the white thread becomes distinct to you" as a psychophysical detection test on simulated absolute sky radiance, with the simulation supplied by a validated backward Monte Carlo radiative transfer model of the twilight sky (described and validated in a companion methods paper, hereafter Paper I). The criterion is relative by construction, comparing tonight's dawn band against tonight's night sky, which gives it two properties no absolute convention has: systematic radiometric errors largely cancel, and the criterion remains defined at high latitude, where the sky brightens without ever becoming fully dark.

Section 2 reviews the textual and observational constraints the criterion must satisfy. Section 3 formulates the detection model. Section 4 describes its single calibrated constant and the inversion study that tests the constant's universality. Section 5 presents the validation against sixteen campaigns, including the out-of-sample Birmingham panel year. Section 6 treats the false dawn. Section 7 quantifies the environmental response: clouds, aerosol, artificial skyglow, and the propagated uncertainty of each computed time. Section 8 discusses jurisprudential scope and limitations.

2. Textual and observational constraints

The criterion is constrained on three sides. Textually: the event is a distinctness threshold ("tabayyana," becomes distinct) between the lit band and the adjacent night sky, judged by an unaided human observer ("lakum," to you); the true dawn is extended laterally ("mustatir(an)") while the false dawn is a narrow vertical column ("mustatil(an)," the wolf's tail); and the evening window closes with the disappearance of shafaq, red or white by school.

Observationally: modern campaigns supply the geometry and the depression statistics. The twilight arch at the moment of first distinctness sits at approximately $2.66 \pm 0.23°$ of apparent altitude [10]; the true dawn band spans roughly 30 to 40° of azimuth at detection [7, 9]; the false dawn's zodiacal wedge is about 20° wide at its base [13]. Depression statistics under dark desert skies: $14.6 \pm 0.3°$ (naked eye, one year, twice monthly, eight observers;

1); $14.01 \pm 0.32^\circ$ over $\sim$80 mornings [9]; $14.90 \pm 0.17^\circ$ by calibrated camera [10]; and, remarkably, a photographic value of 14.25° for the first color difference from Aswan in winter 1908 [11].

Psychophysically: laboratory contrast thresholds [2, 3] predict that the dawn band's photometric excess over the night sky crosses detection threshold near depression 17 to 18°, several degrees before any field campaign reports naked-eye detection. The discrepancy is not a contradiction; it is the known gap between detecting a sharp-edged laboratory stimulus and noticing a borderless luminance gradient spread over tens of degrees of sky, compounded by field conditions (imperfect dark adaptation, unknown fixation, criterion conservatism). Photometers and long-exposure cameras do detect the earlier excess, which is precisely why instrumental campaigns report systematically earlier (deeper) values than naked-eye panels observing the same sky. Any credible model of the textual criterion must bridge this gap explicitly rather than absorb it into retuned physics.

3. The detection criterion

For each scanned solar depression the model simulates seven sky patches at an apparent altitude of 3° (the scan-grid altitude nearest the measured arch altitude of Section 2): five "band" patches spanning $\pm 18^\circ$ of azimuth about the solar azimuth (the mustatir extent), and two "reference" patches at $\pm 100^\circ$, representing the night sky against which distinctness is judged. Each patch's radiance is the sum of the Monte Carlo twilight field (Paper I), the direction-dependent celestial background (zodiacal light, integrated starlight, airglow, moonlight), and artificial skyglow, converted to mesopic luminance from its simulated spectrum.

Detection is a Weber contrast test with the structure of the ayah (the verse). Let $L_i(\theta)$ be the mesopic luminance of band patch i at depression θ, and let B_i be that same patch's deep-night baseline (its own value in full night, so that the celestial background subtracts patch by patch). The patch's signal is its growth above its own baseline, $\Delta L_i(\theta) = L_i(\theta) - B_i$, and it is judged against the observer's adaptation, set by the reference-sky luminance L_{ref}, through the threshold-versus-intensity function: patch i is distinct when

$$\Delta L_i(\theta) \ \geq \ f \cdot C_{\text{thr}}(L_{\text{ref}})\, L_{\text{ref}}, \tag{1}$$

where C_{thr} is the laboratory contrast threshold at the prevailing adaptation, represented by Crumey's 2014 analytic formulation of the Blackwell 1946 threshold-versus-intensity data [2, 3] and f is the field factor of Section 4. Fajr al-sadiq is declared at the depression where the test passes simultaneously across the band's lateral extent (the spread condition); passage of the central patches alone, without the spread, is classified as the false dawn (Section 6). We refer to this test, Eq. (1) together with the spread condition, as the khayt (thread) criterion. The evening events mirror the test as disappearances, with the red channel additionally gated at the cone threshold: below approximately 10^{-3} cd m^{-2}, rod vision carries no color, so there is no red shafaq left to lose; this gate matters for the Hanafi/Shafi'i boundary at high latitude.

Because Eq. (1) is a ratio test against the same night's sky, multiplicative systematics (absolute radiometric scale, uniform cloud attenuation, uniform skyglow lift) largely cancel, and the criterion remains defined wherever the sky physically brightens. At 54.8°N midsummer, inside persistent twilight where no fixed-angle convention returns a time at all, the model still returns a defined khayt dawn (02:55 on the example date), illustrating that the relative criterion stays defined where absolute conventions fail; this is a demonstration of definedness, not a ground-truth validation point.

Figure 1 shows the criterion applied to one example night: the three detection margins per scanned depression, with the false dawn reported where the central margin crosses threshold before the spread margin.

Clouds require one further textual decision. The ayah's "to you" makes the criterion observer-relative: a dawn hidden by an eastern cloud bank has not become distinct to the observer beneath it. The model therefore evaluates the criterion through the measured cloud field rather than above it; a regression test pins the expected asymmetries (an eastern bank delays the dawn detection; a western bank advances the evening disappearance, since dimming a fading band extinguishes its distinctness earlier).

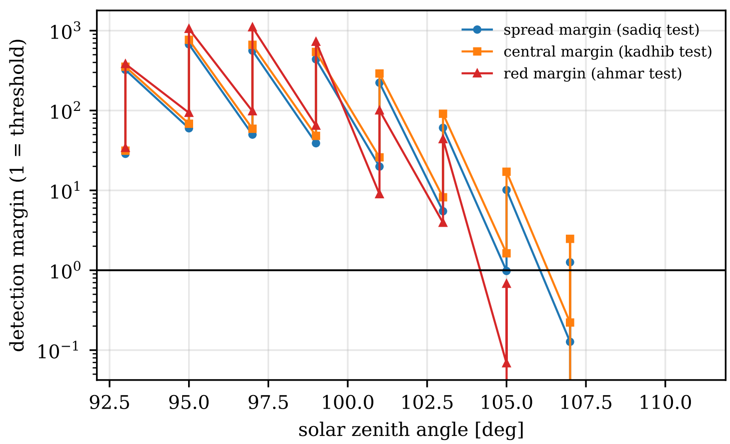

Figure 1: One example night (Mecca, clear sky) is shown: the three detection margins of Eq. (1) against the scanned solar zenith angle. The horizontal line marks the detection threshold; the gap between the central crossing and the spread crossing is the false-dawn interval.

4. Calibration and the universality of the field factor

The single calibrated constant f in Eq. (1) absorbs the laboratory-to-field gap of Section 2. Fixing it requires care, because the published desert campaigns do not report a uniform statistic. The observing groups state their convention explicitly: a bare headline depression is the "highest value of confidence," a mean or median plus one or two standard deviations, not the central tendency [4, 5, 6, 8]. Calibrating to those upper bounds as though they were central means – as an earlier revision did, across the desert body as a whole – biases the target. We therefore fix f only on the campaigns that report a documented central mean, hold it worldwide, and read every other campaign as a test. A decomposition of the calibrated value against the published psychophysics literature (field factor for non-optimal stimuli, blurred-edge degradation, binocular and practiced-observer corrections, visibility-level practice in applied lighting) brackets it comfortably; the decomposition is recorded in the validation archive of the code release.

The constant is pinned by inversion (Figure 2): running a campaign through the engine at a fine factor ladder and solving for the f that reproduces its dawn. Restricted to the campaigns with a documented central mean – Riyadh (14.58°, naked eye), Aswan (14.90°, calibrated camera), and Kottamia (14.665°, naked eye) – the implied factors cluster tightly at 53, 51,

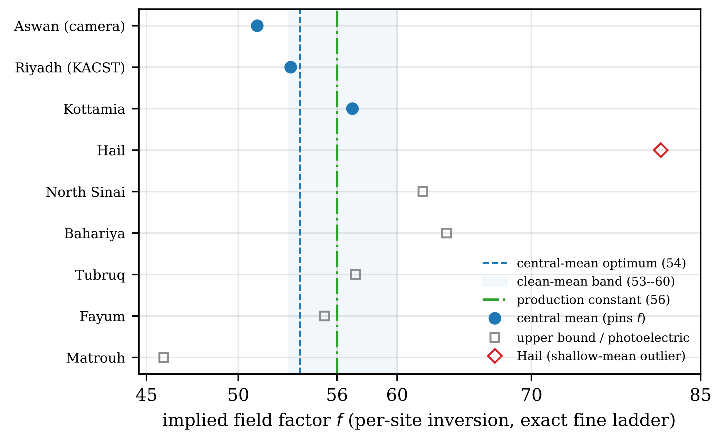

Figure 2: The per-site inversion of the field factor, from exact fine-ladder runs (each point is the f that a campaign implies through the engine). The constant is pinned only by the campaigns with a documented central mean (filled circles): Aswan, Riyadh, and Kottamia imply 51, 53, and 57, an optimum of 53 with a clean-mean plateau to 60 (shaded); the production constant (green, 56) sits within it. Hail (red diamond, implied $f = 81$) is a central-mean outlier, its dawn 14.01° a half-degree shallower than any other clean mean; its leverage alone pulled an earlier three-site fit to 70. The remaining campaigns (open squares) report upper bounds or cross-study photoelectric values, not central means; their implied factors happen to fall in the same band because a site's mean + 2SD dawn is its clear-sky detection limit, which is what the engine predicts. Inverting those campaigns' *raw* central means instead scatters f from 57 to 249 (off scale) and pins nothing.

and 57, with an RMS-optimal $f = 53$ (residual 0.05°), the three agreeing to within four units of f. Hail's central mean, 14.01°, is a half-degree shallower than any other clean mean and implies $f \approx 81$; it is a genuine shallow-dawn outlier, and in an earlier three-site Mecca cluster its leverage alone pulled the fit to 70. Admitting Hail moves the clean optimum only to 60. The production constant is set to $f = 56$, centrally within the 53–60 band that the central-mean campaigns allow, and stable to sub-degree shifts of that basis.

The remaining desert campaigns do not report central means, and must not be read as if they did. Their headline depressions are upper bounds (Tubruq 14.7° = mean + 2SD, Sinai 14.61° = median + σ, Assiut 13.665° = median + 2σ, Wadi Al-Natrun 14.57° = mean + 1SD), while the Bahariya and Matrouh figures come from a separate photoelectric study rather than the naked-eye mean. Their raw central means are far shallower (Tubruq 13.14°, Assiut 11.25°, Matrouh 13.41°), contaminated by all-nights averaging and light pollution that the clear-sky engine does not model; inverting those raw means scatters the implied factor from 57 to 249 and pins nothing. That the engine at $f = 56$ nonetheless reproduces the *upper bounds* is physical, not fitted: a site's mean + 2SD dawn is its deepest, best-visibility, clear-sky detection – exactly the regime the engine predicts. The clean central means and the contaminated campaigns' upper bounds therefore coincide near 14.5–14.9° because both are the clear-sky detection limit, seen two ways. The inversion shows no significant trend with latitude.

The calibration was executed on the frozen final engine by exact GPU runs at a fine factor ladder (the tool and per-site runs are archived with the code release). All campaign numbers in this paper are quoted at $f = 56$; every site outside the three central-mean anchors is a genuine test. The Birmingham out-of-sample statistics are insensitive to the choice within the 53–60 plateau. A later deep-cloud estimator upgrade and an evening-solver bracket correction (companion paper and code history) leave the criterion's dawn detection depressions unchanged to within 0.02° at the calibration anchors, so the calibration is insensitive to both.

5. Validation against observation campaigns

Figure 3 maps the campaign sites, Table 1 summarizes the campaign comparison, and Figure 4 shows the residuals with their published uncertainties. Because eye and instrument measure different events (Section 2), the residual statistics are reported per modality: over the naked-eye campaigns the median absolute residual is 0.28°, and over the instrumental campaigns (SQM knees and calibrated cameras) 0.26°; pooled over all fourteen scored rows it is 0.26°. The constant f is pinned on the three central-mean campaigns (Section 4), and shifting that basis moves it by less than a degree, so each remaining campaign functions as an out-of-sample test rather than a fitted point. For the rows whose published depression is an upper bound (Table 1), the tabulated residual is a lower bound on the mean-basis agreement; it stays small because the engine sits at the clear-sky detection limit those bounds also mark. The Sana'a row is scored only as a consistency check: Sultan's two bracketing events (first glow merged with the zodiacal light at 18.8°; colour divergence at 13.2°) straddle any plausible detection value, so the row cannot discriminate between models and is excluded from the medians. This median of about a quarter-degree of depression is roughly a minute and a

Table 1: Model versus published dawn campaigns (abbreviated; the full table with per-site run brackets is in the repository validation documents). "Engine" is the khayt criterion output matched to the campaign's modality (naked eye versus instrument). "Basis" is the statistic the source reports for its headline depression, which is *not* uniform across campaigns: several are documented upper bounds (mean or median $+k\sigma$, the group's stated "highest value of confidence"), not central means. Residuals are engine minus the published headline value, in degrees of solar depression; for the upper-bound rows this is a lower bound on the mean-basis agreement (the engine sits at the clear-sky detection limit, which coincides with those sites' best-visibility tails). The constant f is pinned on the central-mean campaigns only (Section 4); every row is an out-of-sample test at fixed f.

Site (campaign)	Engine	Observed	Basis	Residual
Riyadh desert (KACST)	14.50	14.6 ± 0.3	mean	-0.10
Hail	14.46	14.01 ± 0.32	mean	$+0.45$
Aswan 2016, camera	14.82	14.90 ± 0.17	camera mean	-0.08
North Sinai	14.16	14.61	med+σ	-0.45
Fayum	14.41	14.8 / SQM 13.75–14.7	range	in range
Matrouh	14.32	14.5	photoelec.	-0.18
Kottamia	14.75	14.66 ± 0.2	mean	$+0.09$
Bahariya	14.63	14.6	photoelec.	$+0.03$
Wadi Al-Natrun	14.00	range 12.48–15.14	mean+1SD	inside
Tubruq (desert horizon)	14.68	14.66–14.7	mean+2SD	0.00
Tubruq (sea horizon)	14.68	13.43–13.48	mean+2SD	$+1.22$
Assiut (agricultural)	14.69	13.665	med+2σ	$+1.03$
Sana'a, 2200 m	15.04	bracket [13.2, 18.8]	bracket	inside
Depok, SQM knee	14.23	14.0 ± 0.6	SQM knee	$+0.23$
Malaysia SQM sites	14.48	14.19 ± 0.52	SQM knee	$+0.29$
Aswan 1909, camera	14.82	14.25	camera	$+0.57$

"mean/med $+k\sigma$" rows are published upper bounds; the sites' central means are shallower (e.g. Tubruq desert 13.14°, Assiut 11.25°) but reflect all-nights or light-polluted averages the clear-sky engine does not model. "photoelec." denotes a cross-study photoelectric value, not the naked-eye mean (Bahariya 13.81°, Matrouh 13.41°). f is pinned on the central-mean campaigns (Riyadh, Aswan, Kottamia), not fitted per row.

half of clock time at mid-latitudes, and comparable to the scatter between the campaigns themselves. Four features deserve comment.

First, the instrument-versus-eye split behaves as the psychophysics requires: the model emits a distinct instrumental (absolute-threshold) output and a naked-eye (khayt) output that differ by about half a degree of depression, and each is compared against data of its own modality. The instrumental output matches the SQM campaigns at the residuals tabulated in

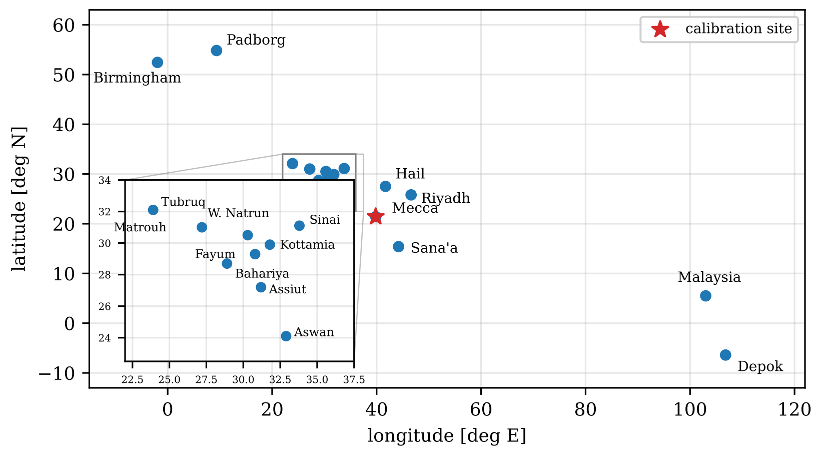

Figure 3: The validation campaign sites are mapped, with the inset expanding the Egyptian and Libyan cluster. Mecca (starred) anchors the criterion historically; the constant is pinned on the central-mean campaigns only (Riyadh, Aswan, Kottamia), so every other campaign is an out-of-sample test.

Table 1 (+0.23° at Depok, +0.29° for the Malaysia sites), while the khayt output matches the visual panels. A 116-year-old photographic campaign [11] is reproduced at +0.57.

Second, one apparent outlier was resolved at its primary source: the Wadi Al-Natrun target circulating in secondary summaries (14.57°) is that paper's mean-plus-one-standard-deviation upper confidence bound, not its campaign mean; the published observed range is 12.48 to 15.14° over 38 mornings, and the model sits inside it. We flag this as a general hazard of campaign meta-analysis: the Egyptian and Saudi campaign literature quotes upper-confidence bounds ("white thread" safety values) alongside means, and a comparison must match the statistic, not the headline number.

Third, the remaining misses are physical rather than statistical, and they are documented as such. The two large residuals sit exactly where the reference-sky assumption breaks: Assiut (+1.03), where the observers faced an agricultural valley whose aerosol and humidity regime the desert climatology cannot represent (the aerosol channel of Section 7 is the identified mechanism, feeding measured optical depth moving the residual back toward the observed range); and the Tubruq sea horizon (+1.22), where the dawn path crosses a directional marine boundary layer that no one-dimensional aerosol column can represent. For the sea-horizon row that mechanism has

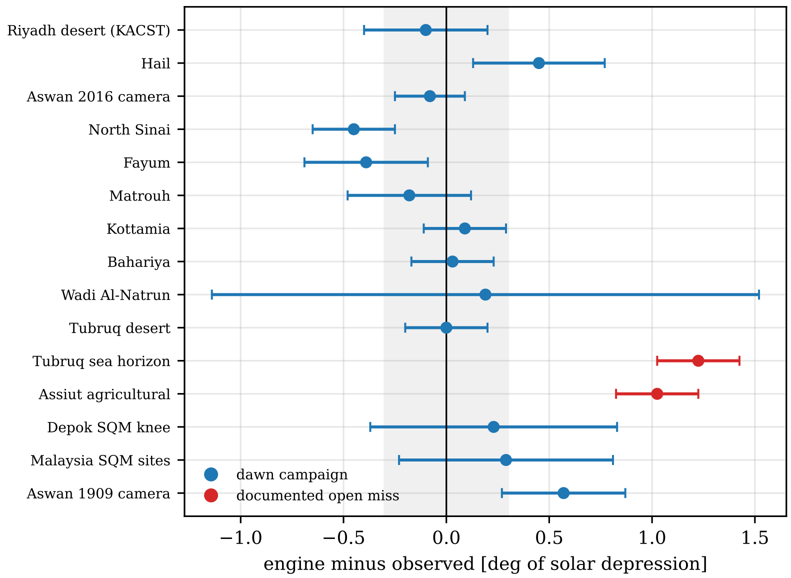

Figure 4: Residuals of Table 1 are shown as a forest plot (engine minus observed, with the campaigns' published uncertainties; the shaded band marks $\pm 0.3°$, close to the pooled median absolute residual of $0.26°$). The two documented open misses, both background-sky physics rather than criterion failures, are discussed in the text.

now been tested directly: representing the marine boundary layer as a georeferenced haze slab confined to the sea cells along the dawn path, at climatological marine optical depth (0.12 at 550 nm, 700 m scale height), shifts the computed sea-horizon dawn by $-0.54°$ of depression averaged over three seasons, closing roughly half the residual (from $+1.2$ to about $+0.5°$), while the same observers' desert horizon remains matched at 0.00. The remaining half is kept as a documented open row rather than absorbed into the calibration; the candidate explanations are marine haze episodically thicker than the climatology, and a different visual task over a featureless sea horizon.

Fourth, the decisive out-of-sample test is Birmingham (Figure 5). The OpenFajr program [12] imaged 300 mornings over a full year at 52.4°N and had a 19-member panel (professional astronomers, timekeeping authorities, and experienced mosque observers) judge 42 clear-horizon date series, producing a double-peaked seasonal curve of dawn depressions between 12.3 and 15.0. The model, calibrated on desert campaigns some twenty degrees of latitude away and never shown these data, reproduces the summer trough (computed 11.9 to 12.6° across the eight June dates after the compressed-

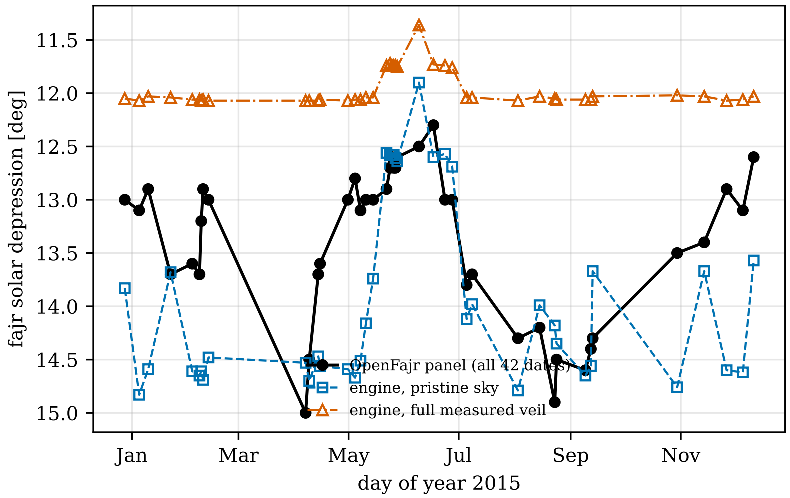

Figure 5: The Birmingham year. OpenFajr consensus-panel dawn depressions (black; all 42 published dates) are compared with the engine under clear sky with zero retuning (blue) and under the full measured atlas veil (purple). Both peaks of the double-peaked curve and the solstice troughs are present in the engine's own seasonal structure (statistics in the text). The winter dates carry the urban veil discussed in Section 7; with the measured skyglow they close to within the reported uncertainty band.

twilight scan refinement of Paper I; panel 12.3 to 12.7) and both peaks of the seasonal curve. The summer relaxation that UK scholars apply as an empirical rule (from $\sim 14.5°$ in winter toward $\sim 12.5°$ in midsummer) emerges from persistent-twilight geometry with no additional freedom. Across all 42 dates the clear-sky statistics are mean $+0.54°$, RMS $0.95°$, $r = 0.51$, and the winter share of the residual is urban-veil physics rather than criterion error: running every winter date under the full measured atlas veil brackets the panel from the other side (mean $-1.13°$, and the panel lies between the clear-sky and full-veil runs on 13 of the 14 winter dates). The bracket width is the quantitative statement of the one input no public feed supplies, the street-light duty cycle at the dawn hour; the panel's position inside it corresponds to a dawn-hour veil of roughly half the atlas's overpass-time value, consistent with documented UK part-night dimming practice (Section 7).

5.1. The evening events

The evening disappearances rest on thinner data than dawn, and the comparison is correspondingly shorter (Table 2). The white-twilight end lies

Table 2: Evening (isha) validation. The white event is matched to instrumental twilight-end data; the red event to the visual literature and Sultan's observation.

Event	Engine (Mecca)	Independent value
Shafaq al-abyad (white end)	17.33°	SQM end $17.99 \pm 0.16°$; muwaqqit 17°
Shafaq al-ahmar (red end)	14.96°	literature 12 to 15°; Sultan 15°

Table 3: The OpenFajr panel year as a benchmark of methods: residuals of each method against the 42 consensus-panel dates.

Method	Mean [deg]	RMS [deg]	Dates defined
This work, clear-sky model	+0.54	0.95	42 of 42
Fixed 15° (ISNA)	+1.62	1.76	42 of 42
Fixed 18° (MWL)	+4.32	4.38	22 of 42

within 0.66° of the instrumental Mecca-region value (17.33 versus $17.99 \pm 0.16°$) and within 0.33° of the classical muwaqqit convention of 17°. The engine's red-end value (14.96°) sits at the upper bound of the sparse visual literature (12 to 15°), essentially coincident with Sultan's 15°. No color-resolved modern campaign exists; until one does, the red calibration should be read as provisional.

5.2. Against the conventions in use

The same 42 panel dates measure the methods the world actually uses (Table 3). The Muslim World League angle of 18° does not exist on 20 of the 42 dates (the Sun never reaches that depression at 52.4°N in summer) and misses by more than four degrees, roughly half an hour of clock time, where it does exist. The 15° convention exists everywhere but misses by 1.76° RMS. The physics-based model is defined on every date of the year at every latitude. Over all 42 dates it is a factor of two more accurate than the best year-round fixed angle (ISNA 15°, 1.76° against the model's 0.95° RMS); the most widely used angle (18°) is undefined on 20 of the 42 dates and misses by 4.4° RMS on the 22 where it exists.

6. The false dawn

Because the criterion evaluates the band patches individually, the false dawn requires no separate machinery: when the central patches pass the

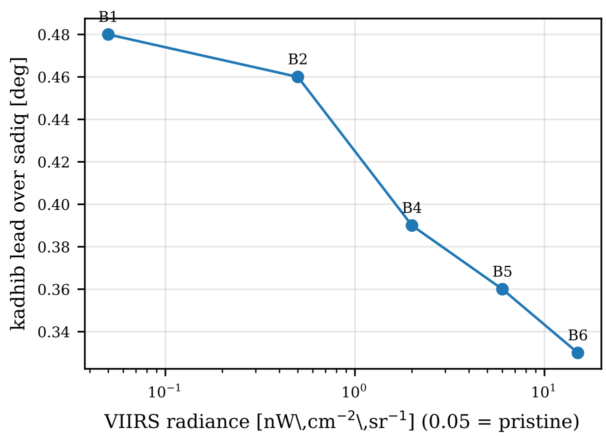

Figure 6: Computed suppression of the false dawn's distinctness by artificial skyglow. The lead of the kadhib detection over the sadiq is shown, in degrees of depression, as the veil is swept from a clear sky to Bortle class 6 at fixed geometry. The monotone compression reproduces the qualitative visibility-literature expectation on a phenomenon that the calibration never used.

distinctness test while the spread condition still fails, the model reports fajr al-kadhib, the narrow vertical brightening of the zodiacal column. At the calibration sites the computed kadhib precedes the sadiq by roughly 0.3 to 0.7° of depression (one and a half to three and a half minutes, varying with site and season), consistent with the classical description of a brief interval in which eating remains permitted while the cautious wait.

The zodiacal character of the event provides an independent test on which the constant was never tuned: the visibility literature places the loss of the zodiacal light near Bortle class 4 to 5 skies. When the artificial veil is swept at fixed geometry, the computed kadhib lead over the sadiq compresses monotonically from 0.48° under a clear sky to 0.33° at Bortle 6 (Figure 6), reproducing the qualitative suppression; the model does not reproduce a hard extinction of the event, which we state as a partial confirmation, since the computed kadhib occurs immediately ahead of dawn, where the wedge is brightest, a configuration the free-standing-cone literature does not directly address.

7. Environmental response and reported uncertainty

Because the criterion is evaluated on tonight's simulated sky, the model quantifies how the times respond to measured conditions, and it reports a propagated uncertainty with every time.

Clouds. A measured three-dimensional cloud field (satellite-reconstructed; Paper I) shifts the dawn detection with the azimuthal asymmetry the observer-relative reading requires: an eastern stratus bank delays fajr while leaving isha nearly untouched, and a western bank advances the evening disappearance. Uniform thin cirrus principally dims the band and delays detection by minutes; the criterion's ratio structure cancels much of a uniform deck's effect, which is why overcast mornings remain computable at all.

Aerosol. The model ingests the Copernicus Atmosphere Monitoring Service (CAMS) aerosol optical depth at the prayer hour as an excess over a baseline measured at the desert calibration campaigns (chosen so the clear-desert body itself carries zero excess by construction). The measured response is 13 to 17 degrees of depression per unit excess optical depth in the linear regime, with the saturation behavior the crossing slope predicts. A sensitivity run at the shipped constant confirms the direction and scale: a continental-average aerosol load representative of the agricultural regime reduces the predicted Assiut depression by of order 1.5°, enough to carry the residual across the observed value, while an urban load overcorrects, so the measured optical depth is bracketed between the clean desert baseline and a continental one. The desert calibration sites carry near-zero measured excess by construction and are therefore essentially unmoved, the differential response expected when one site sits in an aerosol regime the desert baseline does not carry.

Moonlight. The Moon enters the criterion on both sides: moonlight raises the band patches and the reference patches together, and because the test is a ratio against the same night's sky, a bright Moon principally raises the adaptation level (through the threshold-versus-intensity function) rather than shifting the detection wholesale. The lunar state is taken from the ephemeris (true phase angle, parallax, and distance), and the OpenFajr data provide an implicit end-to-end check: the panel reported no lunar structure in their year of detections, and none of the model's residuals against the 42 dates correlates with lunar phase.

Artificial light. The veil is taken from the propagated world atlas, cross-checked against Visible Infrared Imaging Radiometer Suite (VIIRS)

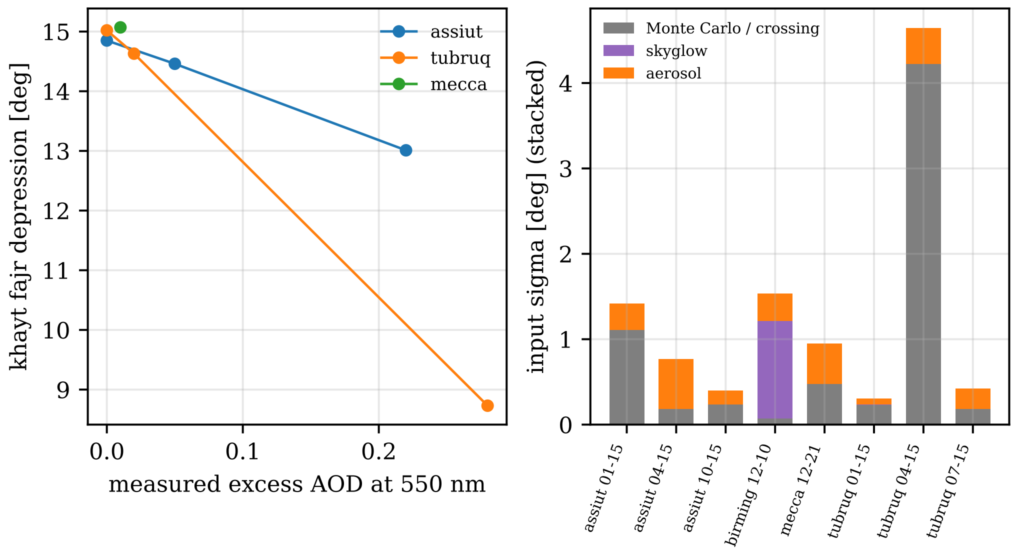

Figure 7: The left panel shows the khayt dawn depression versus the measured excess aerosol optical depth for the validation sites (each point is one campaign date run with the CAMS value at the prayer hour). The right panel shows the propagated uncertainty of each run split by source; the Birmingham winter row is dominated by the skyglow term, itself dominated by the street-light duty cycle.

nighttime lights, and applied in mesopic photometry computed from the source spectrum: a sodium-dominated city veils threshold vision at barely a third of its photopic luminance, an LED city at nearly all of it, a distinction worth more than a factor of two at the adaptation levels of dawn observation. Where part-night switching makes the veil bimodal, the model widens the reported uncertainty accordingly.

Figure 7 shows the measured-aerosol response and the per-run uncertainty decomposition.

Reported uncertainty. Each computed time carries a standard deviation combining Monte Carlo noise, the atlas calibration, the duty-cycle term, and the aerosol input, each propagated through the measured sensitivity of the detection margin at the crossing. Representative values: Mecca ±3.0 minutes; a clear coastal site ±1.8; Birmingham in winter ±7.2, dominated by the duty-cycle term. We regard the width of the urban winter band as a finding in itself: it is a quantitative statement of how much a calendar can currently know.

8. Discussion

Jurisprudential scope. The model computes the physical observable that the sources describe; it does not adjudicate between legal schools. Where the schools differ, the model serves each on its own terms rather than dictating between them: for the isha boundary it tracks the red and the white disappearances as separate physical events (shafaq al-ahmar for the majority Maliki, Shafi'i, and Hanbali position; shafaq al-abyad for the Hanafi school), reporting both times on every night, and the physically forced gap between them (the red band vanishes at the cone threshold while the white persists) gives each school an objective observable matched to its own definition, and its high-latitude behavior (times defined by relative brightening) is offered as physical input to a juristic discussion, not as a ruling. Computed times are explicitly experimental for worship and should be cross-checked against established local calendars; the appropriate near-term use is as an instrument: for calendar committees evaluating their conventions against physics, for campaign design, and for quantifying the environmental corrections (cloud, aerosol, skyglow) that no fixed angle can carry.

Operational accessibility. Because the model's accuracy rests on measured inputs, its practical use by a mosque committee or observatory does not require them: every dynamic input degrades to a documented static fallback (OPAC climatological aerosol by region, the atlas value or a Bortle class for skyglow, clear sky when no cloud product is available, the built-in solar ephemeris), and the computed time then simply carries a wider reported uncertainty, so a committee can run the model offline from coordinates and a date alone and see, from the widths, exactly which measured input would tighten their calendar most. Precomputing an annual table at climatological inputs, as conventional timetables do, is a single command.

Relation to prior computational work. Prior quantitative treatments of fajr visibility have combined twilight brightness models of the Rozenberg lineage with threshold curves [7], or fit depression statistics directly. The present contribution is, to our knowledge, the first to compute the criterion from absolute spectral radiative transfer validated against reference codes and measured skies (Paper I), the first to carry measured three-dimensional cloud fields, measured aerosol, and spectrally resolved artificial light into the detection, and the first to report per-event propagated uncertainties.

Boundaries of the present validation. The evening red event rests on the weakest data (no color-resolved modern campaign exists; the calibration leans on instrumental values and on Sultan's bracketing observations). The sea-horizon problem is open. The field factor, though cross-site invariant, is a single scalar summarizing a rich psychophysical literature; a purpose-built field campaign with controlled adaptation and modern threshold methodology would either confirm it or replace it with something better, and the release includes the instrument protocol for exactly that campaign. Finally, the transport itself carries the documented deep-cloud statistical limits of Paper I.

9. Conclusions

The Quranic dawn criterion is a well-posed detection problem, and solving it as one reproduces more than a century of dawn observation with a median absolute residual of 0.26° of depression (pooled over all scored campaigns) from a single calibrated constant whose per-site inversion shows a cross-site spread the size of the campaigns' own noise with no latitude trend. A fixed angle is the correct first-order summary of dark-desert dawn (the campaigns cluster near 14 to 15°), and the model explains both why that summary works where it works and how it fails where it fails: with season and latitude (Birmingham's curve), with aerosol regime (Assiut), with background sky (Tubruq's two horizons), and with urban light (the winter veil). What replaces the angle is not another constant but a computation, carrying measured inputs and explicitly propagated uncertainties, offered to observers, calendar committees, and jurists as an instrument for the discussion that remains theirs.

Appendix A. Birmingham per-date comparison

The winter rows of Table A.4 carry the site's urban veil. At the recalibrated constant the clear-sky 42-date statistics are mean +0.54°, RMS 0.95°; the full-veil column brackets the panel from the other side, and the azimuthally resolved veil (slant-path Garstang on the measured VIIRS radiance field, Section 7) narrows the bracket from the veiled side (January 24: 11.82 versus the panel's 12.9; December 10: 11.79 versus 12.9). The residual separating the directional-veil runs from the panel is the street-light duty cycle at the dawn hour, the one input no public feed supplies; the June

Table A.4: All 42 OpenFajr panel dates at the recalibrated constant. Panel values are the 19-member consensus depressions; the clear-sky residual is engine (clear sky) minus panel; the final column is the engine under the full measured atlas veil, which together with the clear-sky column brackets the panel (Section 5).

Date 2015	Panel	Clear sky	Residual	Full veil
Jan 11	13.0	13.83	+0.83	12.05
Jan 19	13.1	14.83	+1.73	12.07
Jan 24	12.9	14.59	+1.69	12.03
Feb 06	13.7	13.68	−0.02	12.04
Feb 18	13.6	14.61	+1.01	12.06
Feb 22	13.7	14.65	+0.95	12.06
Feb 23	13.2	14.61	+1.41	12.07
Feb 24	12.9	14.69	+1.79	12.06
Feb 27	13.0	14.48	+1.48	12.07
Apr 20	15.0	14.53	−0.47	12.07
Apr 22	14.5	14.70	+0.20	12.07
Apr 27	13.7	14.47	+0.77	12.07
Apr 28	13.6	14.56	+0.96	12.06
May 13	13.0	14.59	+1.59	12.07
May 17	12.8	14.67	+1.87	12.06
May 20	13.1	14.51	+1.41	12.06
May 23	13.0	14.16	+1.16	12.04
May 27	13.0	13.74	+0.74	12.04
Jun 04	12.9	12.56	−0.34	11.74
Jun 06	12.7	12.58	−0.12	11.72
Jun 07	12.6	12.58	−0.02	11.74
Jun 08	12.7	12.58	−0.12	11.74
Jun 09	12.7	12.61	−0.09	11.74
Jun 10	12.6	12.64	+0.04	11.75
Jun 22	12.5	11.90	−0.60	11.36
Jun 30	12.3	12.60	+0.30	11.73
Jul 06	13.0	12.57	−0.43	11.74
Jul 10	13.0	12.69	−0.31	11.76
Jul 18	13.8	14.12	+0.32	12.04
Jul 21	13.7	13.98	+0.28	12.04
Aug 16	14.3	14.79	+0.49	12.07
Aug 28	14.2	13.99	−0.21	12.03
Sep 06	14.9	14.18	−0.72	12.05
Sep 07	14.5	14.35	−0.15	12.06
Sep 23	14.6	14.65	+0.05	12.06
Sep 26	14.4	14.56	+0.16	12.06
Sep 27	14.3	13.67	−0.63	12.03
Nov 13	13.5	14.76	+1.26	12.02
Nov 28	13.4	13.67	+0.27	12.03
Dec 10	12.9	14.60	+1.70	12.07
Dec 19	13.1	14.62	+1.52	12.06
Dec 25	12.6	13.57	+0.97	12.03

dates are compared clear-sky because their 01:22 GMT dawns fall inside the documented lights-dimmed window.

Code and data availability

The simulator and the detection layer are available at https://github.com/Muno459/twilight (MIT/Apache-2.0); release v1.0.0 corresponds to this paper. The release includes a validation archive containing the campaign run records and the exact data behind every figure and table, together with the scripts that regenerate them. The companion methods paper (Paper I) documents the radiative transfer validation.

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This work stands on the campaigns cited in Table 1: observers who rose before dawn, month after month, from Aswan in 1908 to Birmingham in 2015, and published what they saw.

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